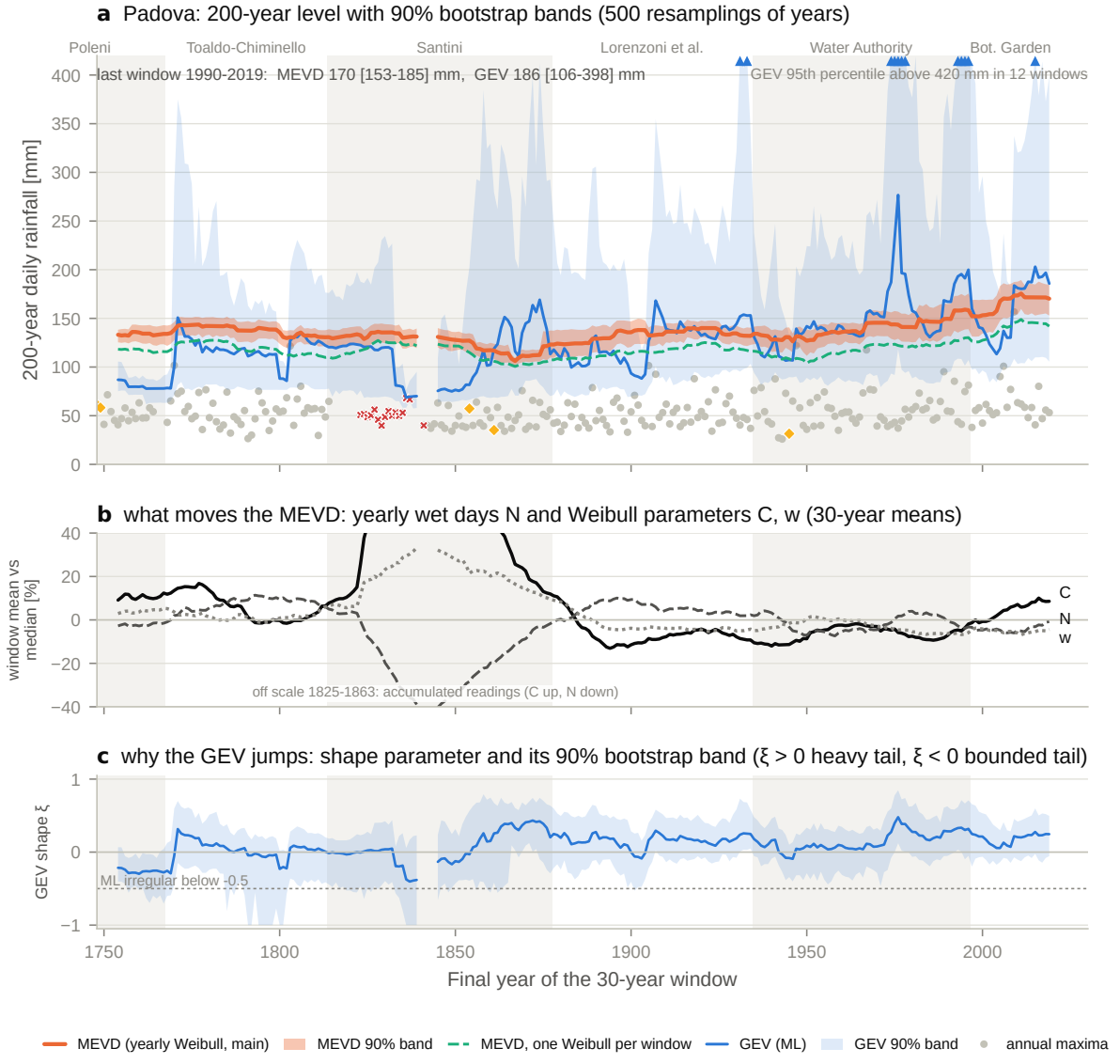
a Padova: 200-year level with 90% bootstrap bands (500 resamplings of years)
Poleni
Toaldo-Chiminello
Santini
Lorenzoni et al.
Water Authority
Bot. Garden
last window 1990-2019: MEVD 170 [153-185] mm, GEV 186 [106-398] mm
GEV 95th percentile above 420 mm in 12 windows
200-year daily rainfall [mm]
400
350
300
250
200
150
100
50
0
b what moves the MEVD: yearly wet days N and Weibull parameters C, w (30-year means)
window mean vs median [%]
40
20
0
−20
−40
C
N
w
off scale 1825-1863: accumulated readings (C up, N down)
c why the GEV jumps: shape parameter and its 90% bootstrap band (ξ > 0 heavy tail, ξ < 0 bounded tail)
GEV shape ξ
1
0
−1
ML irregular below -0.5
1750
1800
1850
1900
1950
2000
Final year of the 30-year window
MEVD (yearly Weibull, main)
MEVD 90% band
MEVD, one Weibull per window
GEV (ML)
GEV 90% band
annual maxima